Reckoning on Recoveries

Dublin Data Science

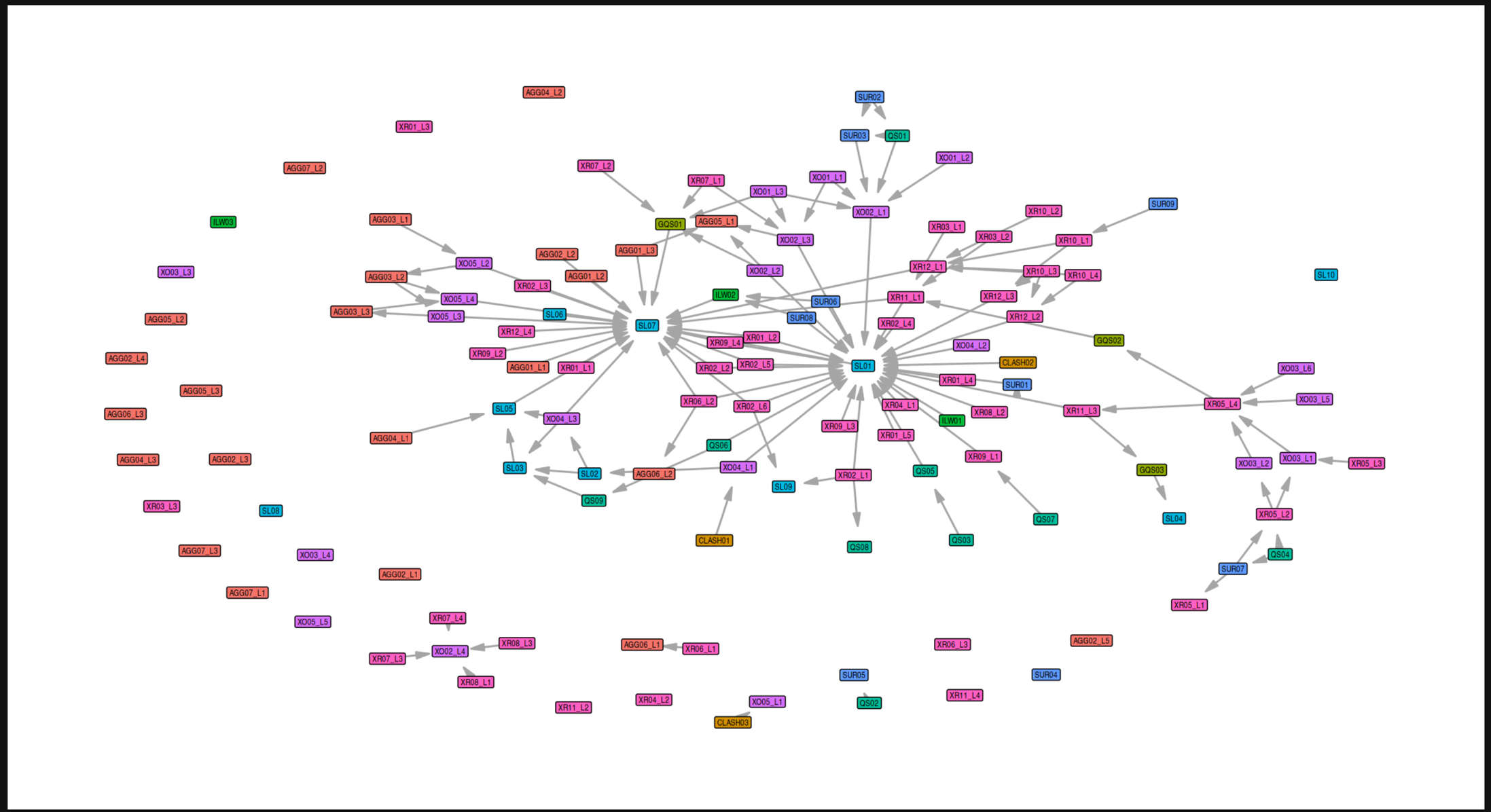
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The Problem



Photo: Miami Herald

Who pays what?


AGG04_L2
SUR02
XR01_L3
SUR03
QS01
AGG07_L2
XR07_L2
XO01_L2
XR07_L1
XO01_L1
XO01_L3
XO02_L1
SUR09
ILW03
AGG03_L1
GQS01
AGG05_L1
XR10_L2
XR03_L1
XO02_L3
XR03_L2
XR10_L1
AGG01_L3
AGG02_L2
XO05_L2
XO02_L2
XR12_L1
XR10_L3
XR10_L4
SL10
XO03_L3
AGG03_L2
AGG01_L2
XR02_L3
XO05_L4
ILW02
XR11_L1
XR12_L3
AGG05_L2
AGG03_L3
XO05_L3
SL06
SUR06
SUR08
XR12_L2
SL07
XR12_L4
XR02_L4
GQS02
XR09_L4
XR01_L2
XO04_L2
AGG02_L4
XR09_L2
AGG01_L1
XR01_L1
XR02_L2
XR02_L5
SL01
CLASH02
XO03_L6
AGG05_L3
XR01_L4
SUR01
XR06_L2
XR02_L6
XR04_L1
XR05_L4
XO03_L5
AGG06_L3
SL05
XO04_L3
XR09_L3
ILW01
XR08_L2
XR11_L3
AGG04_L1
XR01_L5
QS06
XR09_L1
XO03_L2
XO03_L1
XR05_L3
AGG04_L3
AGG02_L3
SL03
SL02
AGG06_L2
XO04_L1
XR02_L1
QS05
GQS03
SL09
XR03_L3
SL08
QS09
QS07
SL04
XR05_L2
AGG07_L3
XO03_L4
CLASH01
QS08
QS03
QS04
AGG02_L1
SUR07
AGG07_L1
XR05_L1
XO05_L5
XR07_L4
AGG02_L5
XR08_L3
AGG06_L1
XR06_L1
XR06_L3
XR07_L3
XO02_L4
SUR05
SUR04
XR08_L1
XR11_L4
XR04_L2
XO05_L1
QS02
XR11_L2
CLASH03

127 treaties

133 inuring edges

The calculation later uses a validated table

How to allocate losses across treaties?

event_id	description	gross_loss
EV01	Small attritional loss (Tower 1 LOB)	$18,668,971
EV02	Large loss, exhausts Tower 1 lower layers	$233,362,133
EV03	Large loss on Tower 3 LOB (through sandwich)	$210,025,919
EV04	Multi-LOB catastrophe -> clash hub	$1,283,491,730
EV05	Mega-cat -> exhausts clash, into agg/ILW/stop loss	$4,200,518,389
EV06	Small attritional loss (Tower 2 LOB)	$14,001,728

Reinsurance Basics

Cedants

transfer premium/risk to

Reinsurers

Capacity

Volatility

Capital

$$\text{Gross Loss} = \text{Ceded} + \text{Retained}$$

Quota Share (QS)

Proportional

Share of premium and losses

25% Quota Share

Gross loss	Ceded (25% QS)	Net retained
2.0m	0.5m	1.5m
10.0m	2.5m	7.5m
40.0m	10.0m	30.0m

Excess of Loss (XoL)

Non-proportional: pays only above *attachment*

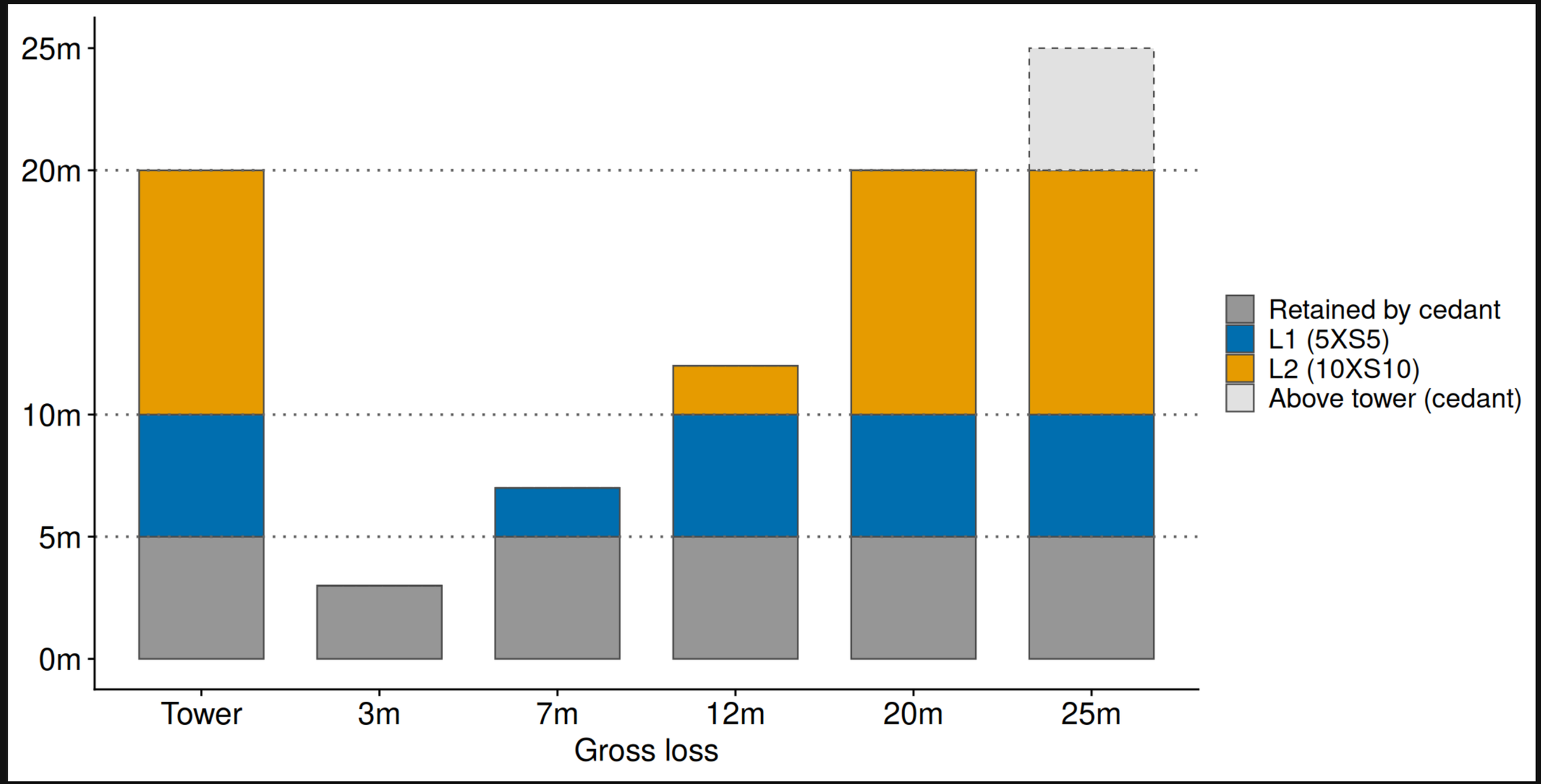
Paid upfront

Limit XS *Attach*

14m Loss

Layer	Band	Recovery
Retained	Below 5.0m	5.0m
L1	5XS5	5.0m
L2	10XS10	4.0m

Who pays at each loss size



Limits and Reinstatements

Occurrence limit: single event

Aggregate limit: all events

Reinstatement: limit restored, usually for extra premium

$$\text{Capacity} = \text{limit} \times (1 + \text{reinstatements})$$

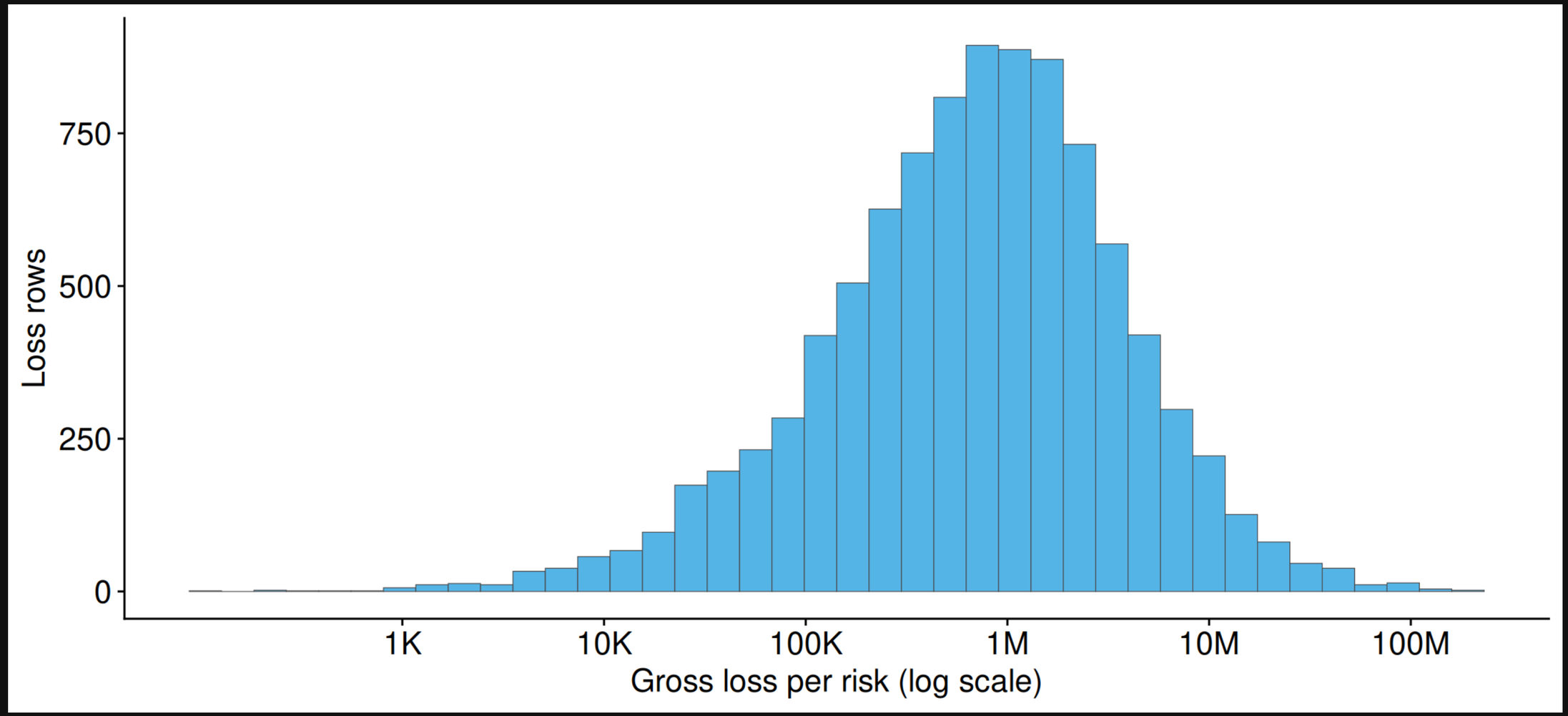
Data

Losses

Loss Class

Loss Class	Premium	Expected Loss
Casualty Auto 2.1 | Europe	$63,617,020	$45,317,233
Casualty Auto 2.1 | North America	$101,208,896	$72,095,598
Casualty Auto 2.1 | UK & Ireland	$34,700,193	$24,718,491
Casualty Liability 2.1 | Asia Pacific	$124,842,890	$95,885,592
Casualty Liability 2.1 | Europe	$171,658,974	$131,842,689
Casualty Liability 2.2 | Asia Pacific	$44,498,823	$21,474,961
Casualty Liability 2.2 | North America	$97,341,175	$46,976,478
Casualty Liability 5.3 | Asia Pacific	$40,803,676	$26,368,341
Casualty Liability 5.3 | Europe	$56,105,054	$36,256,469
Casualty Liability 5.3 | North America	$89,258,040	$57,680,747

Orders of Magnitude



Treaty Data

Treaty ID	Type	Name	Attach	Limit	Cession
trty_024ed8ce01f24896	xol_per_risk	Casualty Liability (P001 2.2) 15XS2	4915777	2106762	NA
trty_02c54572a09f0e38	xol_per_risk	Property Noncat (P001 1.2) 27XS14	47708657	20446567	NA
trty_0864f0beb4716a39	surplus	Casualty Auto (P001 2.1) SP 18m	NA	139368504	0.2578156
trty_0a5a9a3fc02c3472	aggregate_xl	P001 Segment 2.1 27XS20	332073663	15393514	NA
trty_0b8ad8c995ac045b	xol_per_risk	Casualty Liability (P001 2.1) 17XS6	28141937	12060830	NA
trty_0beba38395c53493	xol_per_occurrence	Property Cat (P001 3.2) 8XS4	45217721	19379023	NA
trty_0c41f5ff3fe0f25c	aggregate_xl	Marine Cargo (P001 4.1) 3XS2	65695831	11549762	NA
trty_0d1ef0a1fc51b903	stop_loss	Portfolio 001 38XS27	2473127184	746948731	NA
trty_0d9a3817bf374c70	xol_per_occurrence	Property Cat (P001 3.2) 8XS5	92281063	39549027	NA
trty_0ff6b5afcce49180	xol_per_risk	Property Noncat (P001 1.1) 3XS1	23647271	10134545	NA

Treaty Coverage

Treaty	Class
trty_024ed8ce01f24896	Casualty Liability 2.2 | North America
trty_02c54572a09f0e38	Property Noncat 1.2 | Asia Pacific
trty_02c54572a09f0e38	Property Noncat 1.2 | Latin America
trty_02c54572a09f0e38	Property Noncat 1.2 | North America
trty_0864f0beb4716a39	Casualty Auto 2.1 | North America
trty_0864f0beb4716a39	Casualty Auto 2.1 | UK & Ireland
trty_0a5a9a3fc02c3472	Casualty Auto 2.1 | UK & Ireland
trty_0a5a9a3fc02c3472	Property Cat 2.1 | UK & Ireland
trty_0b8ad8c995ac045b	Casualty Liability 2.1 | Asia Pacific
trty_0b8ad8c995ac045b	Casualty Liability 2.1 | Europe

Treaties as Functions

$$\mathit{Treaty} : \text{Losses} \to \text{Recoveries}$$

Recovery Accounting

10XS10 XoL

10 million excess of 10 million

Segment	Gross Loss
Construction	5.0M
Legal	10.0M
I.T.	5.0M
Total	20.0M

Treaty recovers 10M

How do we allocate them?

Segment	Treaty	Recovery
Construction	10XS10	2.5M
Legal	10XS10	5.0M
I.T.	10XS10	2.5M
Total	10XS10	10.0M

25% QS

25% quota-share (often with a limit)

Segment	Gross Loss
Construction	5.0M
Legal	10.0M
I.T.	5.0M
Total	20.0M

Treaty recovers 5M

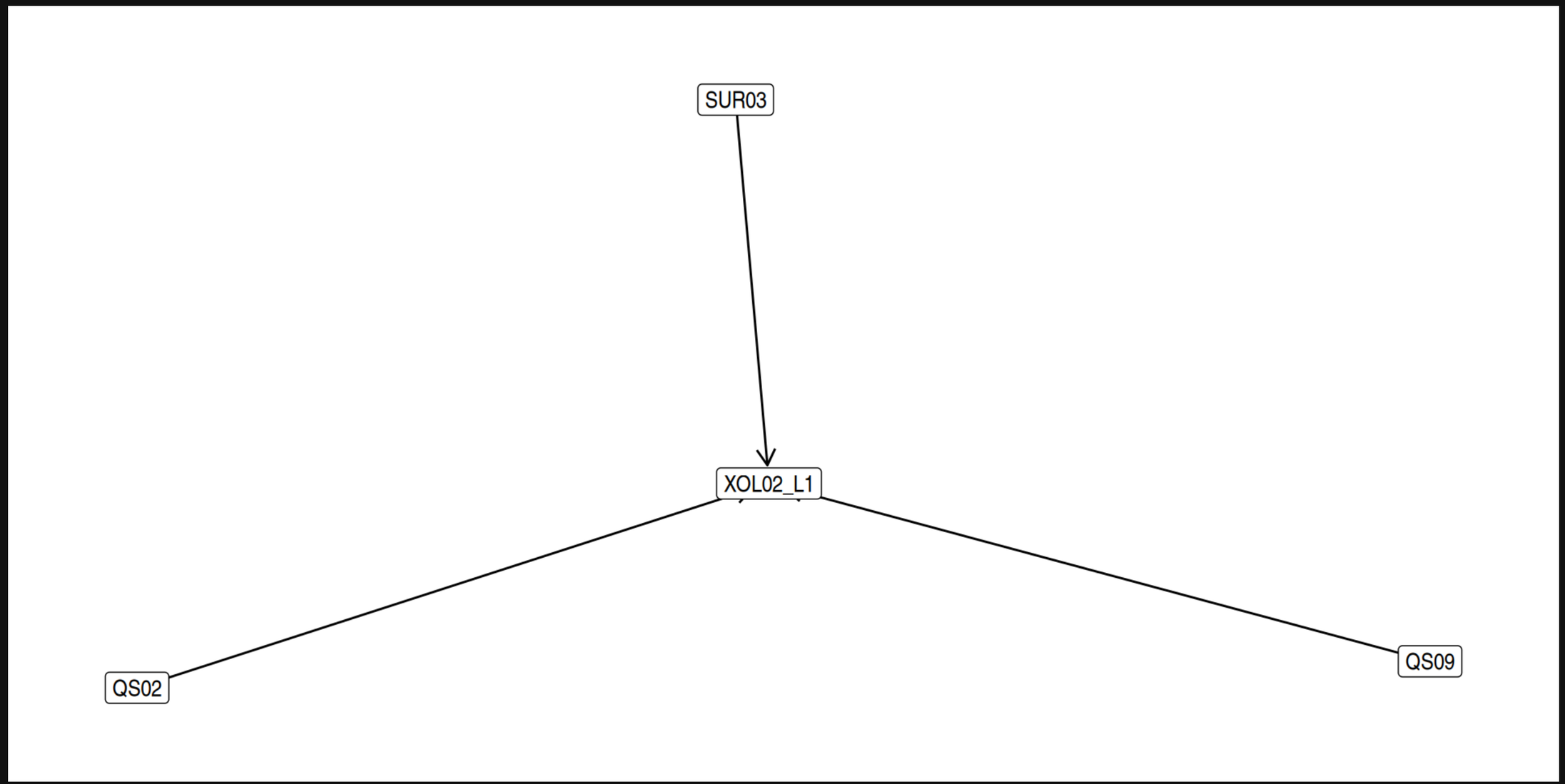
Segment	Treaty	Recovery
Construction	25% QS	1.25M
Legal	25% QS	2.50M
I.T.	25% QS	1.25M
Total	25% QS	5.00M

Inuring

Sequencing of treaties

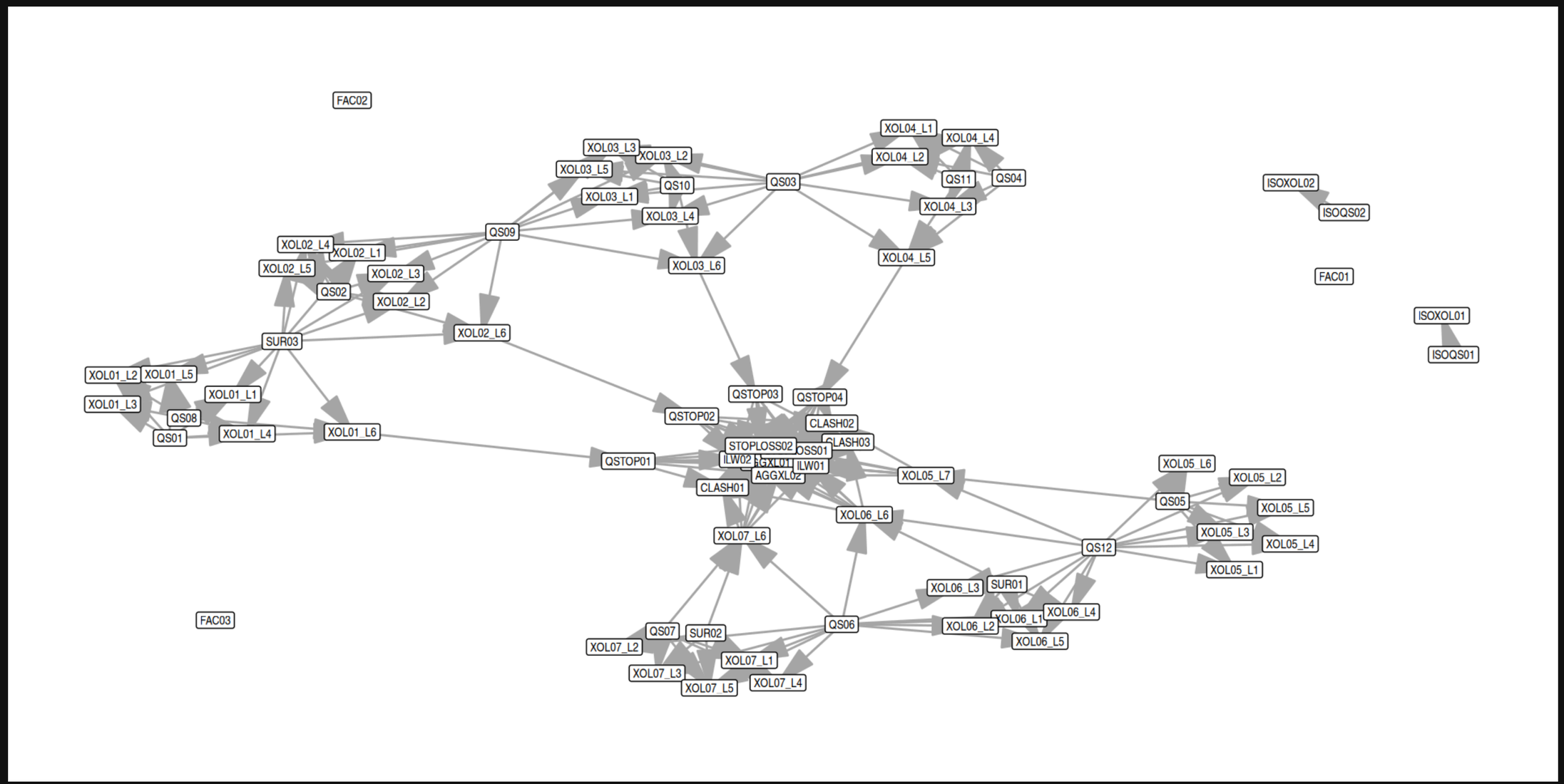
Gross / Subject Losses

Subject Loss = Gross Loss - Inuring Recoveries


SUR03
XOL02_L1
QS02
QS09

Topological Sort

Inuring graph dictates calculation order


FAC02
XOL04_L1
XOL04_L4
XOL03_L3
XOL03_L2
XOL04_L2
XOL03_L5
QS10
QS03
QS11
QS04
XOL03_L1
XOL04_L3
XOL03_L4
QS09
XOL02_L4
XOL02_L1
XOL04_L5
XOL03_L6
XOL02_L5
XOL02_L3
QS02
XOL02_L2
XOL02_L6
SUR03
XOL01_L2
XOL01_L5
XOL01_L1
XOL01_L3
QS08
QS01
XOL01_L4
XOL01_L6
QSTOP03
QSTOP04
QSTOP02
CLASH02
CLASH03
STOPLOSS02
QSTOP01
ILW02
ILW01
AGGXL02
CLASH01
XOL05_L7
XOL06_L6
XOL07_L6
FAC03
QS07
SUR02
XOL07_L2
XOL07_L1
XOL07_L3
XOL07_L5
XOL07_L4
QS06
XOL06_L3
SUR01
XOL06_L1
XOL06_L4
XOL06_L2
XOL06_L5
QS12
XOL05_L6
XOL05_L2
QS05
XOL05_L5
XOL05_L3
XOL05_L4
XOL05_L1
ISOXOL02
ISOQS02
FAC01
ISOXOL01
ISOQS01

calc_order	treaty_id	treaty_type	treaty_label	description
1	QS12	quota_share	Continental Europe GL QS 37%	QS on lob_5ec9f2cf609e0409, 37% ceded
2	QS03	quota_share	Continental Europe GL QS 38%	QS on lob_072509d90dee0973, 38% ceded
3	QS05	quota_share	APAC Cargo Specie QS 43%	QS on lob_7dbc7e4c7c380b86, 43% ceded
4	SUR01	surplus_share	APAC GL Surplus 15%	Surplus Share on lob_5c340bb16d8b1403, 15% ceded
5	SUR02	surplus_share	US Southeast Casualty Umbrella Surplus 27%	Surplus Share on lob_48e5e1a412975a55, 27% ceded
6	QS10	quota_share	UK D&O QS 43%	QS on lob_144138733b9281ac, 43% ceded
7	QS04	quota_share	US Southeast PI QS 38%	QS on lob_6fd96135091380a0, 38% ceded
8	QS09	quota_share	Continental Europe GL QS 22%	QS on lob_5a151f644f09e803, 22% ceded
9	QS07	quota_share	US Southeast Credit QS 32%	QS on lob_31ba4be968fa1714, 32% ceded
10	QS02	quota_share	UK GL QS 41%	QS on lob_60eae97305b9d0bc, 41% ceded
11	QS11	quota_share	Continental Europe General Hull QS 38%	QS on lob_2bf3407d167ed9d1, 38% ceded
12	SUR03	surplus_share	US Southeast Fleet Auto Surplus 16%	Surplus Share on lob_0704378d4b16e6b1, 16% ceded
13	QS01	quota_share	UK Property Non-Cat QS 43%	QS on lob_7945c1570c9592db, 43% ceded
14	QS06	quota_share	Continental Europe Casualty Umbrella QS 44%	QS on lob_6d41f7d04ca5523b, 44% ceded
15	QS08	quota_share	US Southeast Cargo Specie QS 23%	QS on lob_65066c667f902fa2, 23% ceded

Orchestration

1. Construct graph
2. Construct treaty functions
3. Determine treaty/loss coverage
4. Topological sort
5. Loop over each node:

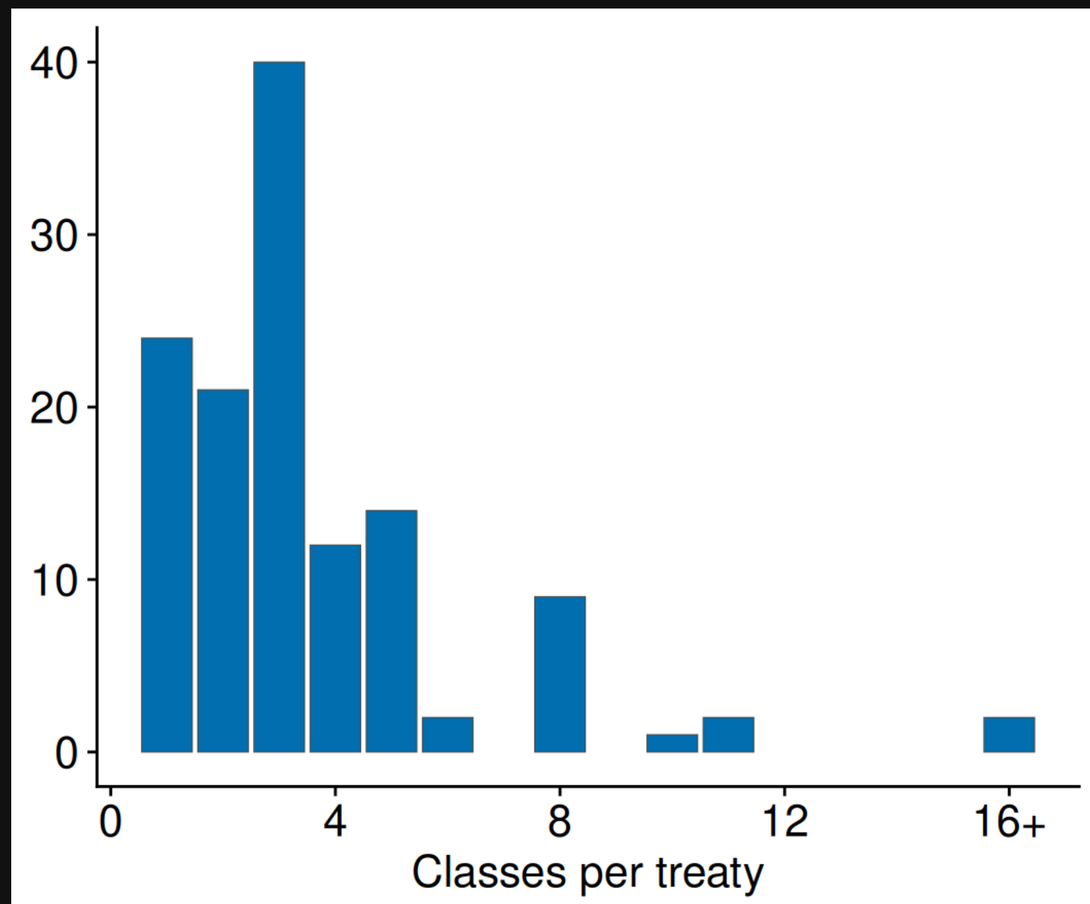
    a. List inuring treaties
    b. List inuring losses/recoveries
    c. Keep covered losses/recoveries
    d. Calculate subject losses
    e. Calculate recoveries

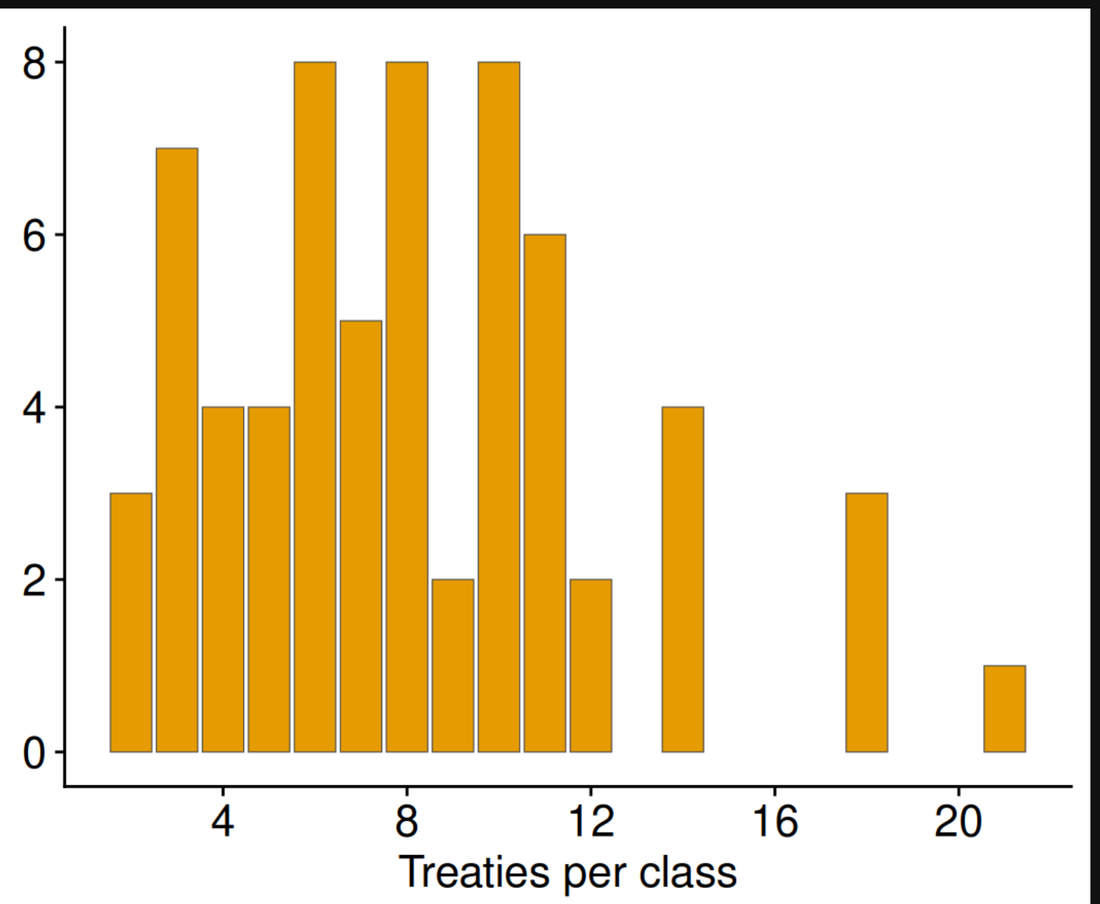
6. Gather all per-node outputs
7. Construct outputs

Treaty / Loss coverage non-trivial

Multiple ways to determine coverage

Many-to-many relationship


40
30
20
10
0
0
4
8
12
16+
Classes per treaty


8
6
4
2
0
4
8
12
16
20
Treaties per class

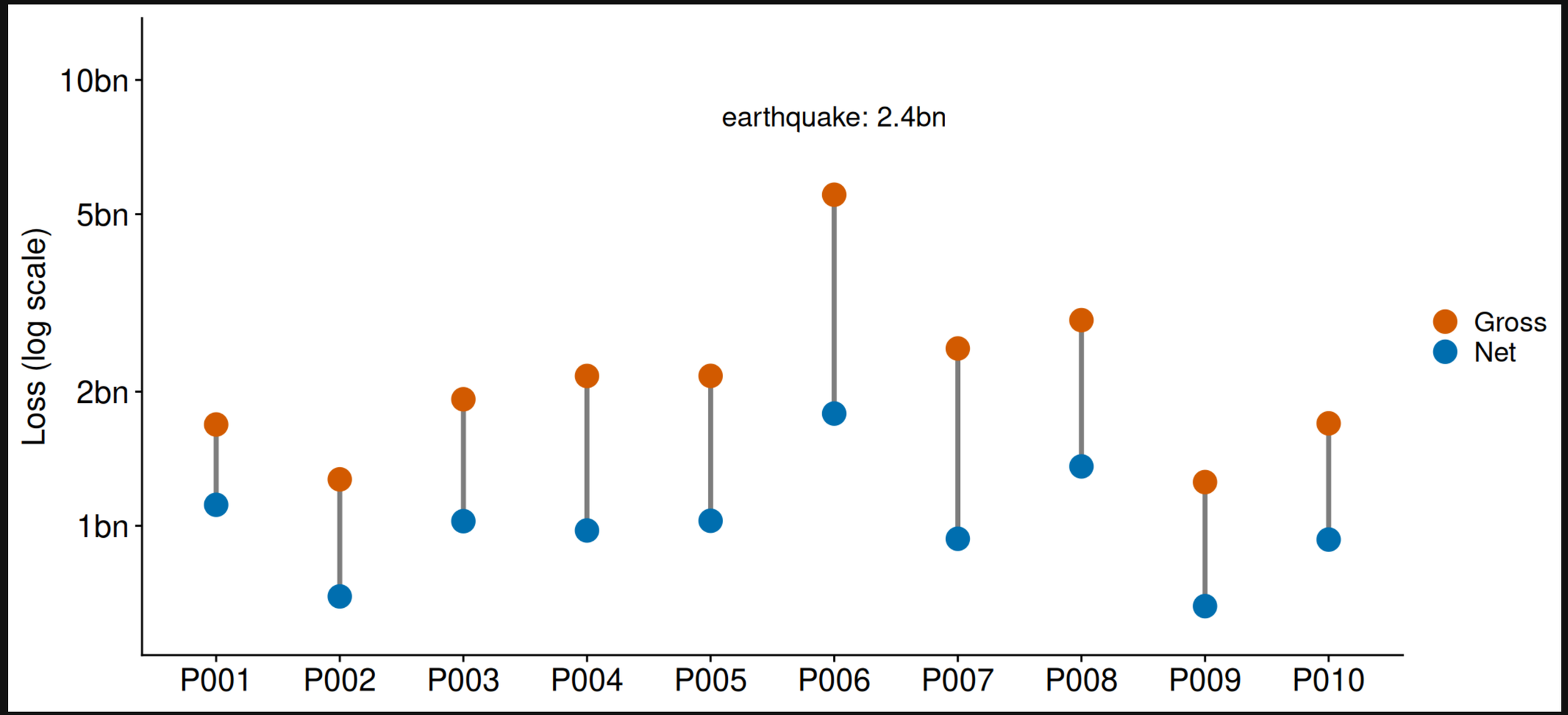
Gross / net premium losses

Organisation unit

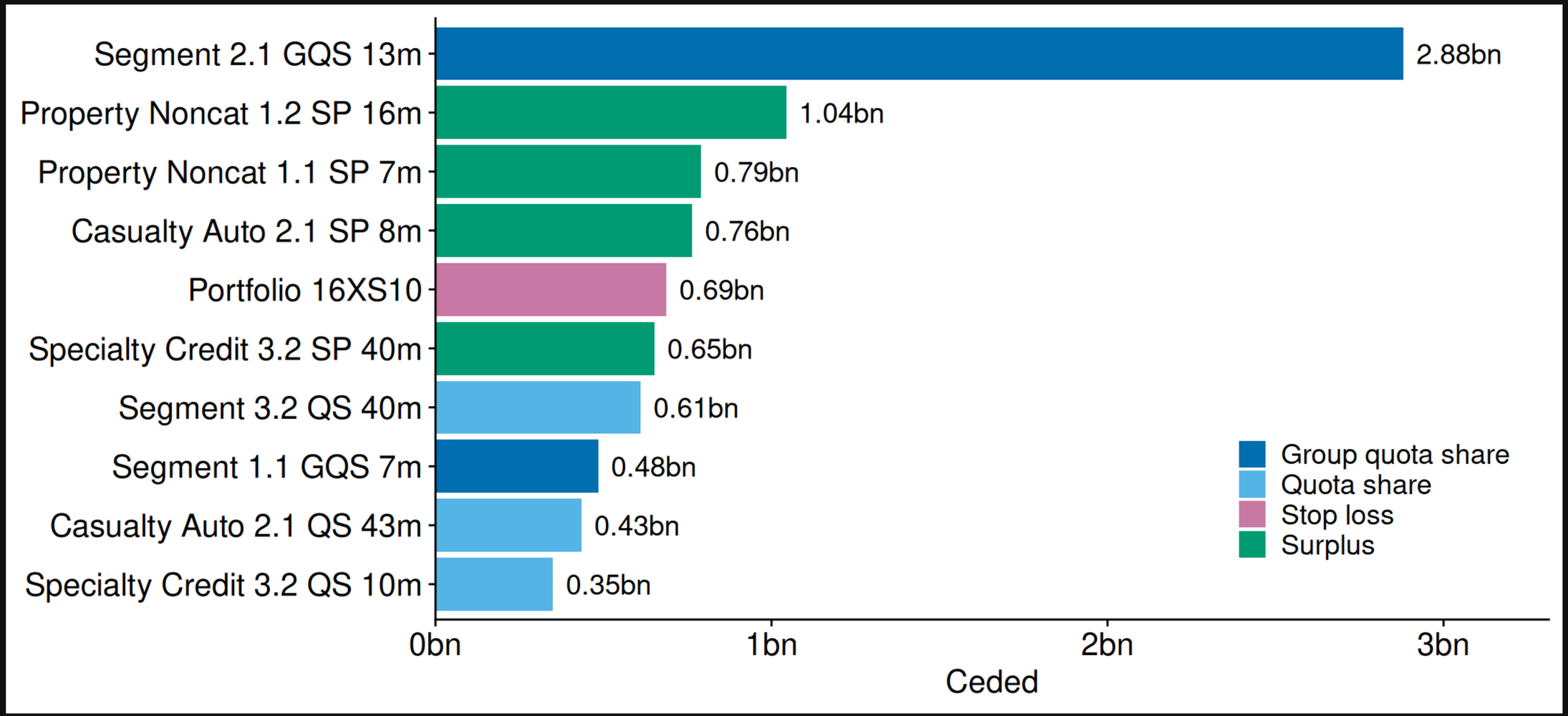
Regulatory authority

Running the Numbers

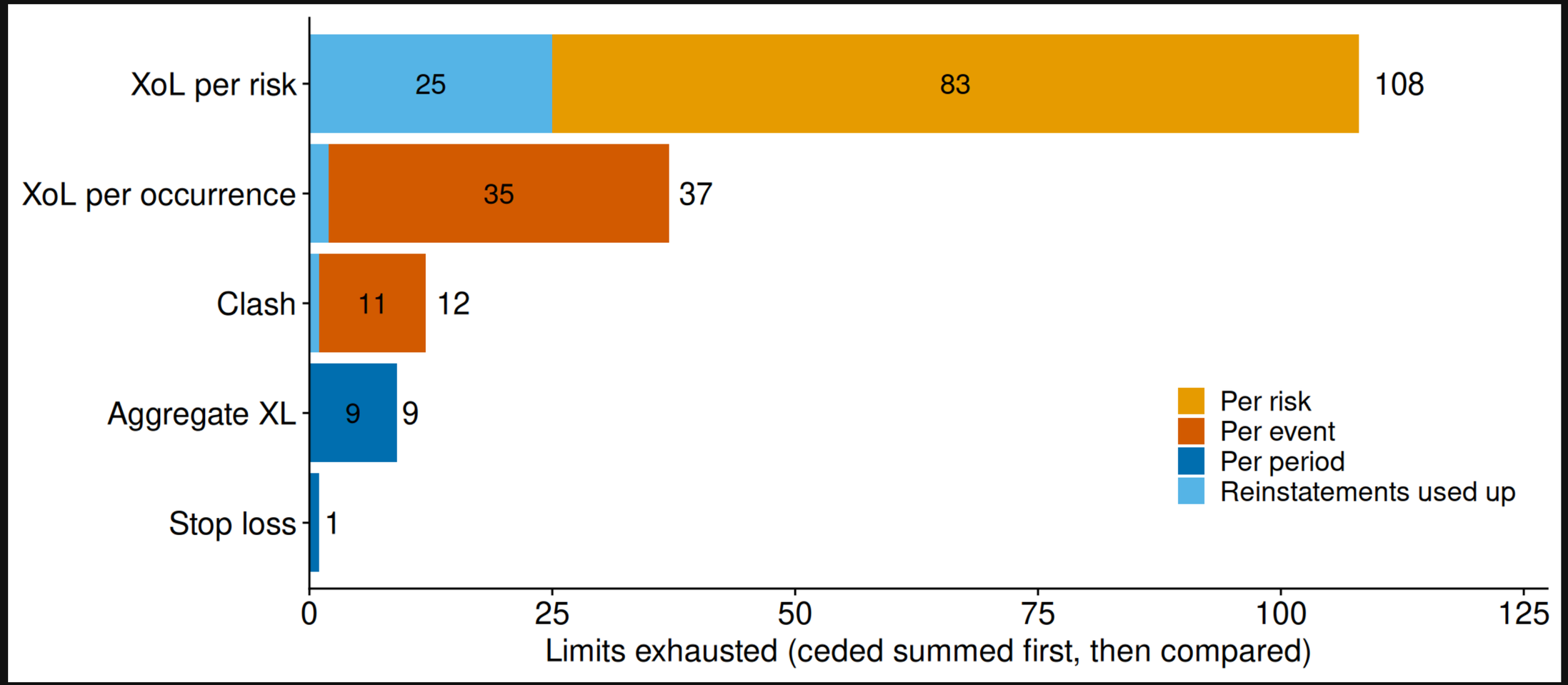
All the Losses, Gross to Net



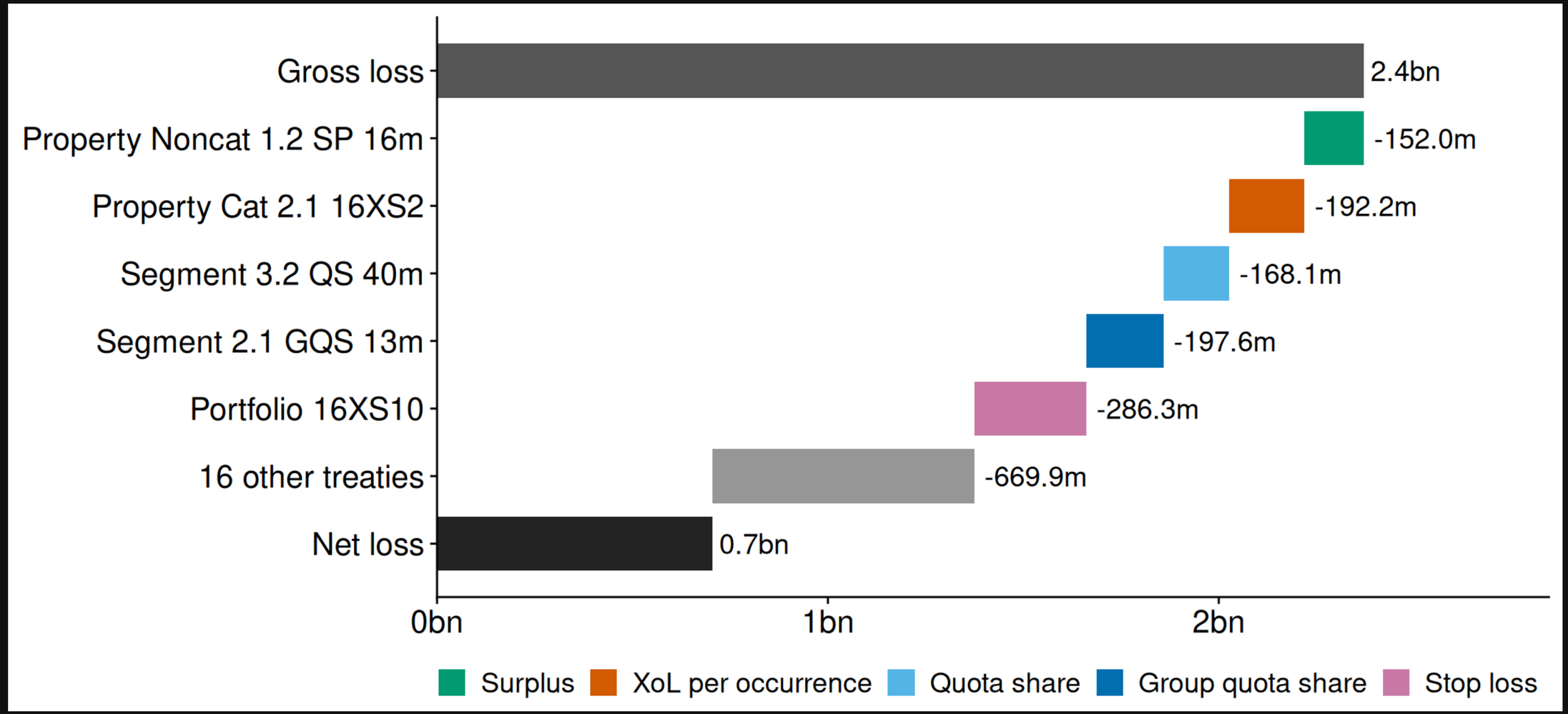
Top 10 Treaties by Ceded



Exhaustion


XoL per risk
XoL per occurrence
Clash
Aggregate XL
Stop loss
25
83
108
35
37
11
12
9
9
1
0
25
50
75
100
125
Limits exhausted (ceded summed first, then compared)
Per risk
Per event
Per period
Reinstatements used up

Loss Apportionment



The Invariants Hold

Check	Result
Gross = ceded + net on every row	PASS
Net ≥ 0 on every row	PASS
Ceded ≤ subject loss ≤ gross	PASS
Limits and reinstatement caps hold	PASS
Same seeds, same ledger	PASS
Any application order, same ledger	PASS

Questions

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https://github.com/kaybenleroll/data_workshops/talk_dds_rei